CNS_ALA-3

CYBER CRIME
CASE STUDIES

Incidents, Impact, Response & Prevention

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A study of five cyber-risk examples across business, public services and national security

Introduction

Cyber crime includes illegal or unauthorised activity carried out through computers, networks, websites and digital accounts. Its effects can include exposure of personal information, financial fraud, service disruption and loss of public trust. This project reviews five examples related to the supplied reference PDF, using a reorganised format and fresh wording. Where the reference describes an allegation or an illustrative news clipping rather than a verified incident, that uncertainty is retained.

Objectives of the Study

- Identify the type of cyber risk represented in each case.
- Summarise the reported event and its possible consequences.
- Review response and recovery considerations.
- Recommend practical controls that can reduce similar risks.
- Distinguish confirmed information from claims that require verification.

1. Equifax: Exposure of Consumer Credit Data



Incident type: Large-scale personal-data breach

Region / period: United States, disclosed in 2017

Case overview

The reference material describes a breach at Equifax, a major credit-reporting agency. Attackers exploited a known weakness in an internet-facing Apache Struts application that had not been patched in time. The company announced the incident in September 2017. Early reporting cited about 143 million U.S. consumers; later investigations put the number at roughly 147 million.

How the attack worked

The key failure was exploitation of an unpatched web-application vulnerability. Once inside, the attackers accessed systems holding sensitive records and extracted information. The supplied reference does not identify phishing as the confirmed entry method.

Impact and response

Information exposed included names, dates of birth, Social Security numbers, addresses and, for some individuals, driver's licence numbers or payment-card details. The breach increased the risk of identity theft and fraudulent credit activity. Equifax brought in outside specialists, notified consumers, offered monitoring services and faced lawsuits and regulatory action.

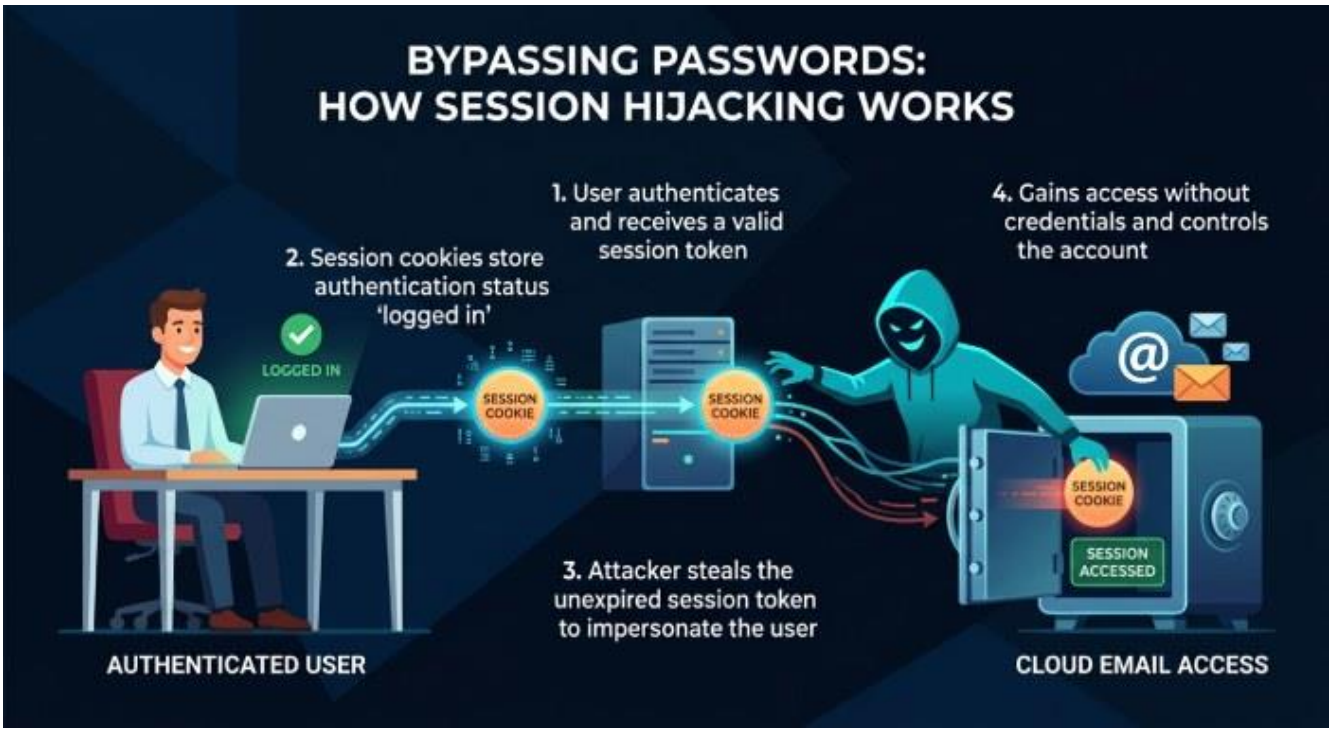
Financial / recovery note: A 2019 FTC-led settlement required at least $575 million in relief and could reach $700 million, depending on claims and conditions. This is not a complete accounting of all recovery costs.

Prevention lessons

- Prioritise and verify critical security patches.
- Keep an inventory of public-facing applications and scan them regularly.
- Separate web services from databases and limit privileges.
- Encrypt sensitive records and monitor unusual access or large data transfers.
- Prepare and practise a response plan before an incident occurs.

Main takeaway: A known vulnerability can become a major breach when patching, asset tracking and detection are all weak.

2. Optus: Customer Information Accessed Without Authorisation



Incident type: Telecommunications customer-data exposure

Region / period: Australia, disclosed in September 2022

Case overview

According to the reference, Optus reported that an unauthorised party had accessed customer information. Public reporting connected the incident with up to around 9.8 million records or people, but the fields exposed differed between individuals. The supplied article also highlights cybersecurity workforce readiness and policy concerns.

Likely risk area described

The reference discusses an exposed application interface and concerns about access controls. It does not establish phishing as the confirmed cause, and exact technical details should be tied to official findings rather than inferred from a headline.

Impact and response

Potential consequences included identity fraud, impersonation and misuse of identity-document details. Optus notified customers and worked with authorities; regulators examined the incident. The event also intensified discussion about data retention and the security of customer-facing interfaces.

Financial / recovery note: The supplied clipping does not establish one definitive total recovery cost. Investigation, customer support, legal exposure and reputational harm may all contribute.

Prevention lessons

- Authenticate and authorise every sensitive API request.
- Restrict access to customer-data endpoints and test for data enumeration.
- Retain only information that is genuinely needed.
- Use rate limits, encryption, monitoring and a rehearsed notification process.
- Assign clear security responsibilities and invest in staff capability.

Main takeaway: Secure technology and organisational readiness must work together.

3. Website Malware and Compromise: Los Angeles Times Context



Los Angeles Times

latimes.com

Malware Attack!

Stop the presses! I've been hacked

By Garland Technology

Hacker, strange man in hoody accessed the LA Times user database by using a Malware attack

Incident type: Malicious code / website compromise (reference image is illustrative)

Region / period: United States; related reporting mentioned in 2018

What the reference supports

The supplied PDF includes a Los Angeles Times-style clipping headed “Malware Attack!” but warns that the image alone does not prove a specific database breach. It separately mentions reporting about cryptocurrency-mining code affecting a Los Angeles Times webpage in 2018. That activity is generally called cryptojacking and is different from confirmed theft of a subscriber database.

Possible attack methods

Website malware can be introduced through vulnerable content-management software, stolen administrator credentials, altered scripts or poorly configured cloud storage. Cryptojacking runs unauthorised mining code in a visitor’s browser. These are examples of possible methods for this threat category, not proof that all occurred in the supplied clipping.

Impact and recovery

Possible effects include slow pages, higher computing costs, service interruption, malicious redirects and reduced reader trust. A responsible response would isolate affected systems, preserve logs, remove unauthorised code, rotate credentials, patch components and check backups before restoring services. The reference provides no reliable specific loss figure or recovery duration.

Prevention lessons

- Update content-management systems, themes and plugins.

- Require multi-factor authentication for administrators.
- Limit cloud permissions and protect deployment secrets.
- Use file-integrity monitoring and content-security policies.
- Monitor abnormal CPU use, outbound connections and unexpected scripts.

Main takeaway: Verify the facts behind an incident report and use layered technical controls to reduce website risk.

4. Online Fraud: Domain Identity and Payment Verification



Incident type: Phishing, impersonation and payment fraud risks

Region / context: India; policy and prevention focus

Case overview

The supplied news clipping discusses stronger electronic KYC checks for domain-name registrants and bank-side beneficiary-name checks. It is a policy-focused report, not evidence of one specific attack or a quantified loss. Online fraud may involve look-alike domains, fake login pages, spoofed messages or payment instructions that imitate a trusted organisation.

How victims may be targeted

A fraudster may register a similar-looking domain, send a deceptive message or pressure a person to reveal credentials or transfer money to an account controlled by a mule. Beneficiary-name checks can flag mismatches but cannot prevent every scam.

Impact and response

Victims may lose funds, expose identity documents or lose control of accounts. The reference reports proposals for better identity verification and payment checks. In a real case, a victim should contact the bank promptly, preserve messages and transaction details, and report through the appropriate cybercrime channel.

Prevention lessons

- Check the spelling and domain of websites before entering information.
- Use phishing-resistant multi-factor authentication where possible.
- Verify changes to payment details through a separate trusted channel.
- Use email-authentication controls such as SPF, DKIM and DMARC.
- Train staff to recognise urgency, secrecy and unusual payment requests.

Main takeaway: Identity checks are useful, but they work best alongside user awareness, fast reporting and cooperation between organisations.

5. Alleged DRDO Documents Advertised on the Dark Web



31GB of DRDO data available for sale on dark web for $8K: Report

Verifying Report's Authenticity, Say Defence Sources

The publicly available sample documents appear to contain restricted technical information purportedly related to approvals or signatures of senior scientists

Incident type: Alleged sensitive-document exposure

Region / context: India; authenticity described as under verification

Case overview

The reference clipping reports an allegation that a collection of documents said to relate to India's Defence Research and Development Organisation was offered for sale online. It also states that authenticity was being checked. A forum listing alone does not prove that documents are genuine, classified, current or stolen.

Possible exposure routes

If sensitive files are exposed, possible routes can include compromised credentials, phishing, malware, an unpatched system, insecure file sharing, misconfigured storage or an insider copying information. The supplied clipping does not confirm which, if any, route occurred.

Impact and response

If restricted defence material were confirmed as exposed, it could reveal research, technical capabilities or internal processes. If the material were fabricated, the claim could still consume investigative resources and harm reputation. The reference describes verification as the initial response and gives no confirmed financial loss or recovery timeline.

Prevention lessons

- Classify files by sensitivity and restrict access by role.
- Encrypt documents in storage and during transfer.
- Require multi-factor authentication and review access logs.
- Monitor bulk downloads and unusual access patterns.
- Audit third-party access and practise incident-response procedures.

Main takeaway: Treat claims as unverified until evidence supports them; sensitive information needs strict access control and monitoring.

Cross-Case Comparison

Case	Primary risk	Key control	Evidence caution
Equifax	Personal-data breach	Patch management and segmentation	Reported figures changed after investigation
Optus	Customer-data exposure	API access control and data minimisation	Exposure varied by person
Website malware	Malicious scripts / cryptojacking	Updates, MFA and monitoring	Clipping does not prove a database theft
Online fraud	Impersonation and payment scams	Domain checks and payment verification	Policy report, not one quantified incident
Alleged DRDO files	Potential sensitive-document exposure	Classification and least privilege	Authenticity was described as unverified

General Recommendations

- Keep software, websites and plugins updated with a documented patch process.
- Use multi-factor authentication and least-privilege access for important accounts.
- Encrypt personal and confidential information and minimise how much is retained.
- Monitor logs, unusual downloads, suspicious sign-ins and abnormal network activity.
- Maintain clean backups and test restoration rather than assuming backups will work.
- Create an incident-response plan that defines responsibilities, evidence handling and notifications.
- Train users to identify phishing, impersonation and suspicious payment requests.
- Communicate carefully: distinguish confirmed findings from allegations or illustrative material.

Conclusion

These case studies show that cyber risk is not limited to one attack method. Weak patching, exposed interfaces, malicious code, social engineering and poor access management can all create serious consequences. Strong prevention combines technical safeguards, clear responsibility, user awareness and a tested response plan. Accurate reporting is equally important: an allegation or illustrative image should not be presented as a confirmed breach without supporting evidence.

Reference Note

This alternative project is based on the topics and qualifications contained in the user-provided PDF, “Cyber Crime Case Studies.” The supplied material references The Mountain Enterprise (15 September 2017) and the U.S. Federal Trade Commission’s 2019 Equifax settlement, a Herald Sun clipping about Optus, a Los Angeles Times-style illustrative clipping concerning malware, an Indian policy-focused news clipping on domain identity and payment checks, and a clipping alleging DRDO-related documents were advertised on the dark web. The source PDF does not provide fully verifiable bibliographic details for every clipping, so uncertain items are described cautiously.